

Show Python code for circle calculations

- format output to x2 decimal places

Planning

Plan your code before you begin typing the code. Begin with a sample calculation (or two) and then use the IPO chart to plan your input, processing, and calculations. Remember that "an hour of coding can save you minutes of planning." (coding humor) ... a few minutes of planning can save you hours of coding!

Sample Calculations

Create sample calculations by hand to plan and test your code. Given specific input, perform the required calculations to determine the expected output.

	radius	area	circumference	output area + circumference
1.	(1)	$\pi * 1 * 1 = 3.14$	$2 * \pi * 1 = 6.28$	
2.	(2)	$\pi * 2 * 2 = 12.57$	$2 * \pi * 2 = 12.57$	
3.	(5)	$\pi * 5 * 5 = 78.54$	$2 * \pi * 5 = 31.42$	
4.	(10)	$\pi * 10 * 10 = 314.16$	$2 * \pi * 10 = 62.83$	

Input Processing Output

What values do you need to begin? What calculations or processing is needed to determine the answer? What results must be shown? What variable names will you use?

Input

radius

Processing

$\pi = 3.14$
area =
 $\pi * radius * 2$

circumference =

$2 * \pi * radius$

Output

circumference

area